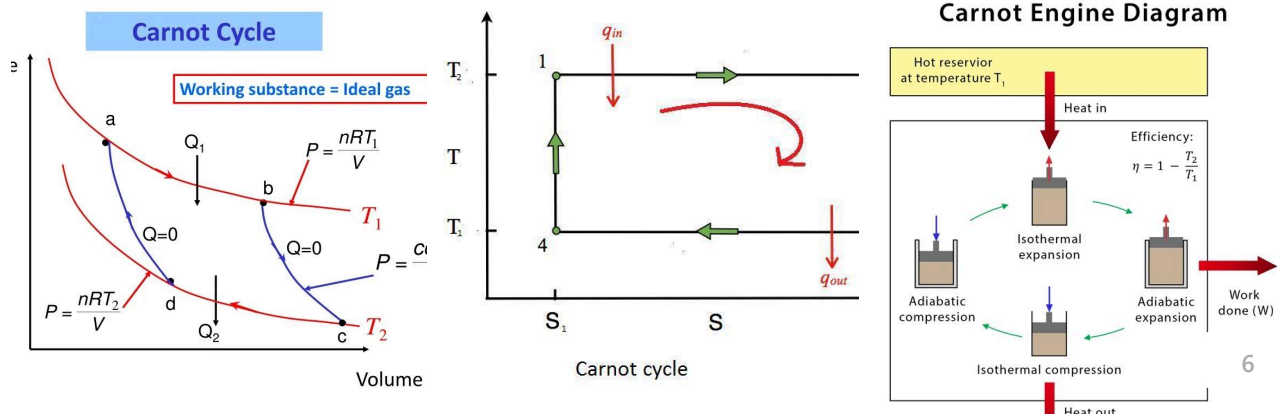


This is a copy of a shared ChatGPT conversation
Report conversation

In thermodynamics, a cycle is a series of processes after which the working fluid returns to its initial state. Important engineering cycles include **Carnot, Rankine, Brayton (Joule), Otto, Diesel, and Dual cycles**.

Below are the P-V, T-S, and h-s diagrams, brief explanations, examples, and differences.

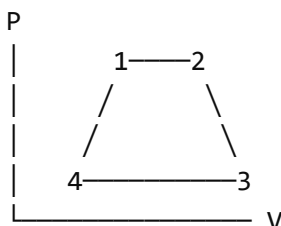
1. Carnot Cycle



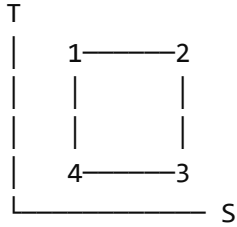
Processes

- 1–2: Isothermal expansion (heat added).
- 2–3: Adiabatic expansion.
- 3–4: Isothermal compression (heat rejected).
- 4–1: Adiabatic compression.

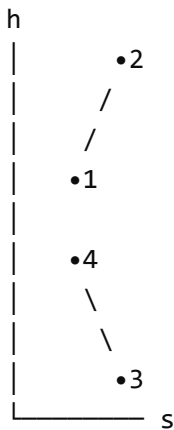
P-V Diagram (schematic)



T-S Diagram



h-s (Mollier) Diagram



Features

- Highest possible efficiency between two temperatures.
- Ideal reversible cycle.
- Not practical for actual power plants.

Example: An ideal heat engine operating between a hot reservoir and a cold reservoir.

2. Rankine Cycle (Steam Power Cycle)

Ideal Rankine Cycle on a p-v diagram

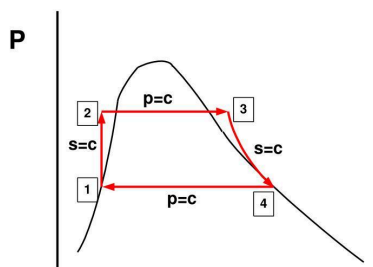
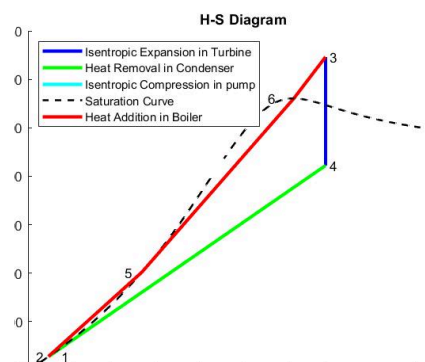
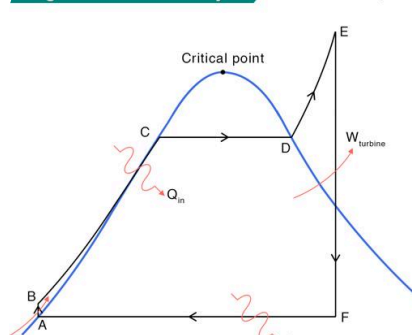


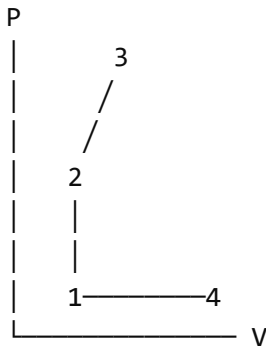
Diagram for Rankine Cycle



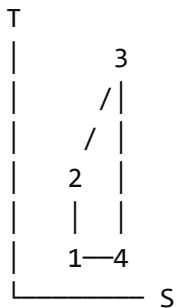
Processes

- 1–2: Pumping of water.
- 2–3: Heat addition in boiler.
- 3–4: Expansion in turbine.
- 4–1: Condensation in condenser.

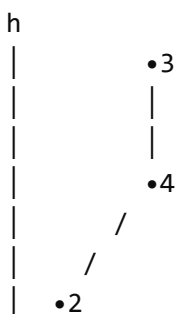
P–V Diagram



T–S Diagram



h–s Diagram



•1

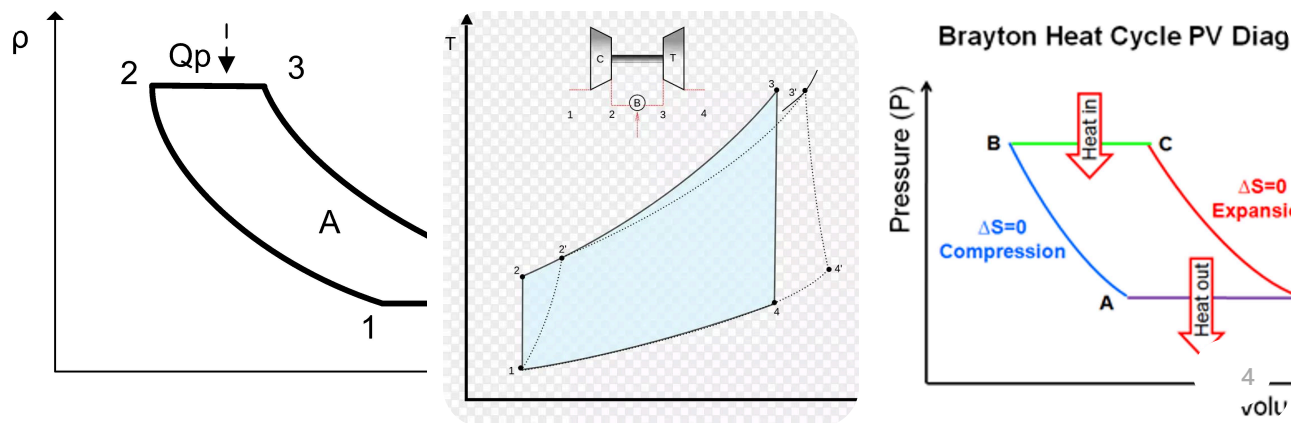
s

Features

- Used in steam power plants.
- Uses water/steam as working fluid.
- Practical and widely used.

Example: Thermal power plant.

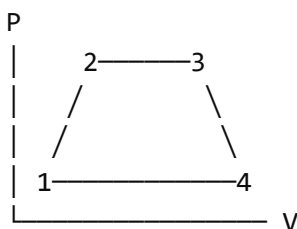
3. Brayton (Joule) Cycle



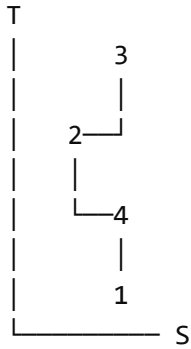
Processes

- 1–2: Isentropic compression.
- 2–3: Constant-pressure heat addition.
- 3–4: Isentropic expansion.
- 4–1: Constant-pressure heat rejection.

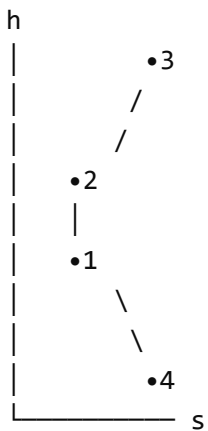
P–V Diagram



T-S Diagram



h-s Diagram

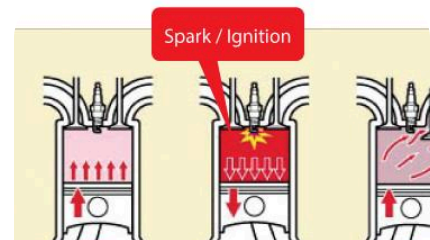
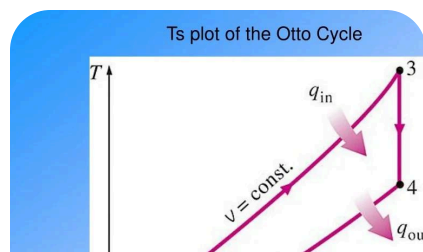
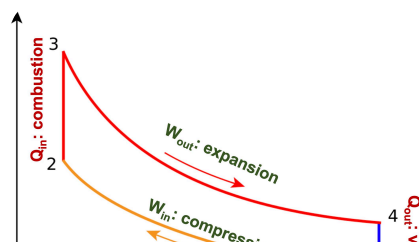


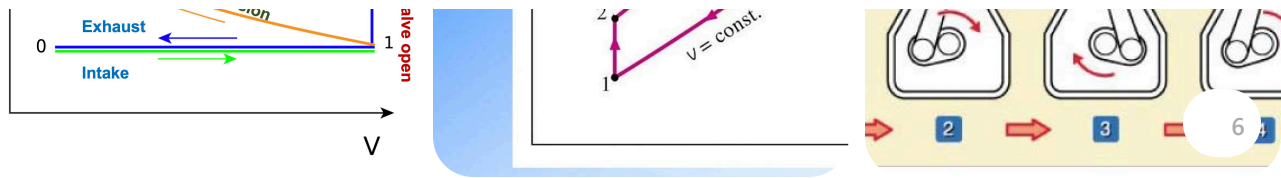
Features

- Works with air/gas.
- Constant-pressure combustion.
- Used in gas turbines.

Example: Aircraft jet engines and gas turbine power plants.

4. Otto Cycle

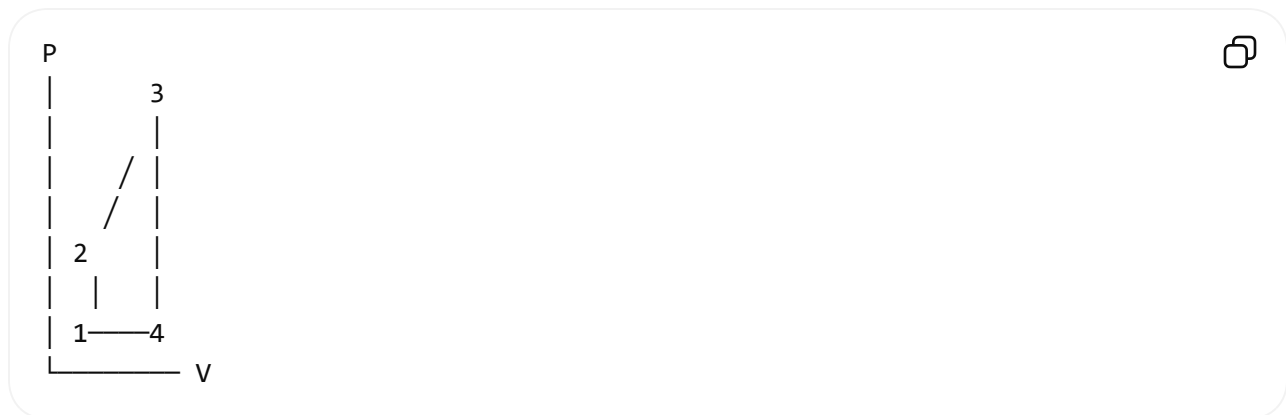




Processes

- Isentropic compression.
- Constant-volume heat addition.
- Isentropic expansion.
- Constant-volume heat rejection.

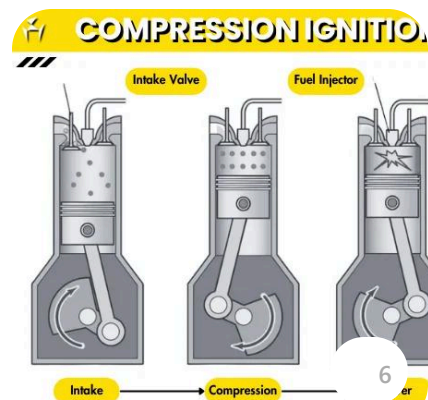
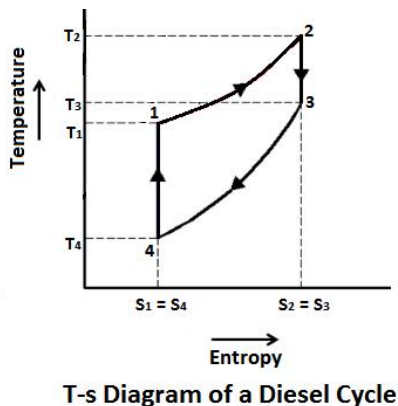
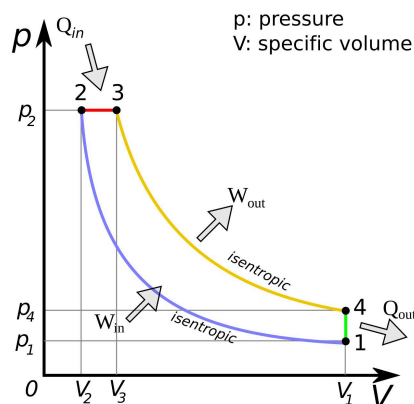
P-V Diagram



Example

Petrol (gasoline) engines.

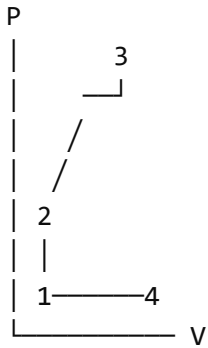
5. Diesel Cycle



Processes

- Isentropic compression.
- **Constant-pressure heat addition.**
- Isentropic expansion.
- Constant-volume heat rejection.

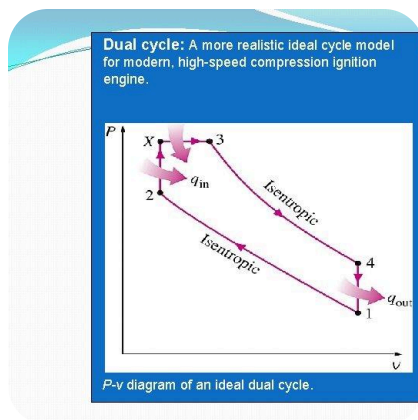
P-V Diagram



Example

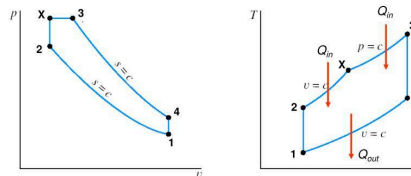
Diesel engines used in trucks and buses.

6. Dual Combustion Cycle

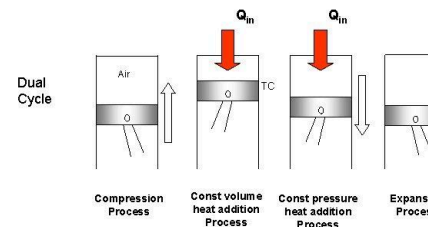


Dual Cycle

- Process 1 → 2 Isentropic compression
- Process 2 → X Constant volume heat addition
- Process X → 3 Constant pressure heat addition
- Process 3 → 4 Isentropic expansion
- Process 4 → 1 Constant volume heat rejection



Thermodynamic Dual Cycle



Processes

- Part of heat added at **constant volume**.
- Remaining heat added at **constant pressure**.
- Followed by isentropic expansion and constant-volume heat rejection.

Example: Modern high-speed diesel engines.

Difference Between Major Cycles

Cycle	Working Fluid	Heat Addition	Application
Carnot	Any ideal fluid	Isothermal	Ideal heat engine
Rankine	Steam	Boiler heating	Steam power plant
Brayton (Joule)	Air/Gas	Constant pressure	Gas turbine
Otto	Air-fuel	Constant volume	Petrol engine
Diesel	Air-fuel	Constant pressure	Diesel engine
Dual	Air-fuel	Constant volume + pressure	Modern diesel engine

Important Diagram Meanings

- **P–V:** Pressure vs Volume. Shows compression and expansion work.
- **T–V:** Temperature vs Volume. Useful for visualizing heating and cooling.
- **T–S:** Temperature vs Entropy. Area under the curve represents heat transfer.
- **h–s (Mollier):** Enthalpy vs Entropy. Commonly used for steam turbines and refrigeration systems.

Quick Memory Trick

- **Carnot:** Maximum theoretical efficiency.
- **Rankine:** Steam power plant.
- **Brayton/Joule:** Gas turbine and jet engine.
- **Otto:** Petrol engine.
- **Diesel:** Diesel engine.
- **Dual:** Combination of Otto and Diesel.

These are the most commonly asked thermodynamic cycles along with their **P–V**, **T–S**, and **h–s diagrams**, applications, and key differences.

